

Simulated Hospitalization Trends, Treatment Persistence, and Care-Pathway Outcomes With Cardionex in Chronic Heart Failure Management

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Abstract

Adults with chronic heart failure often require long-term multidisciplinary management to reduce symptom burden, prevent clinical deterioration, and support adherence to complex treatment regimens. Hospitalization remains a clinically important event, but treatment persistence, outpatient escalation, and care-pathway stability also influence real-world disease-management value. To describe simulated hospitalization trends, treatment persistence, outpatient care utilization, and safety observations among adults with chronic heart failure receiving Cardionex, the drug name associated with Corvanta, as a simulated investigational therapeutic concept. This simulated 52-week global observational journal-style study included adults with chronic heart failure receiving background standard care. Participants initiated once-daily Cardionex within a structured care-pathway simulation. Outcomes included heart-failure hospitalization, urgent outpatient intensification, emergency-care encounters, treatment persistence, adherence, patient-reported symptom stability, and treatment-emergent adverse events. Results were summarized descriptively overall and by baseline risk category. Among 912 simulated participants, mean age was 68.1 years, 44.2% were female, and 46.7% had at least one heart-failure hospitalization in the prior year. At 52 weeks, 14.8% experienced a simulated heart-failure hospitalization, while 23.6% required urgent outpatient therapy intensification. Treatment persistence was 79.9%, and 76.8% maintained adherence of at least 80%. Safety observations were consistent with a medically complex chronic heart-failure population. This simulated study suggests that Cardionex could be evaluated through integrated hospitalization, adherence, and outpatient care-pathway endpoints. Prospective clinical validation would be required.

Keywords

Chronic heart failure; hospitalization; adherence; treatment persistence; outpatient care pathway; Cardionex; simulated observational study

Introduction

Chronic heart failure is associated with recurrent symptom worsening, frequent healthcare utilization, reduced physical capacity, and substantial burden on patients, caregivers, and healthcare systems. Hospitalization for heart failure remains one of the most visible markers of disease instability, but it is not the only clinically meaningful outcome. For many adults, repeated outpatient therapy adjustments, emergency-care visits, diuretic intensification, monitoring burden, and incomplete adherence contribute to declining quality of life and fragmented care.

Cardiologists and internal medicine specialists frequently manage heart failure across longitudinal settings rather than isolated episodes. In this context, a therapeutic strategy should be evaluated not only by symptom or biomarker changes, but also by whether it supports treatment persistence, reduces avoidable care escalation, and integrates into daily disease management. This is particularly important in global populations where care access, monitoring infrastructure, medication affordability, comorbidity burden, and follow-up frequency vary widely.

Corvanta is the brand name for Cardionex, a simulated investigational therapeutic concept for chronic heart failure management in adults. This manuscript presents a journal-style simulated observational study designed to examine hospitalization trends, outpatient care-pathway outcomes, adherence, and safety observations among adults receiving Cardionex in addition to background care. The objective is not to establish efficacy, but to model the type of evidence structure that could support future clinical development and medical-affairs evaluation.

Methods

Study Design

This was a simulated 52-week, prospective, global observational study. The design modeled a chronic heart-failure care pathway in which adults received background standard care and initiated Cardionex as a once-daily oral investigational therapeutic concept. All results are simulated and intended for internal scientific-content workflow demonstration.

Study Population

The simulated cohort included adults with chronic heart failure managed in cardiology and internal medicine settings. Participants were modeled across geographic regions to reflect global care diversity. Risk strata were assigned at baseline based on prior hospitalization, NYHA functional class, kidney disease status, and natriuretic peptide risk profile.

Inclusion Criteria

Participants were eligible if they met the following simulated criteria:

- Age 18 years or older
- Chronic heart failure diagnosis for at least 6 months
- NYHA class II to IV symptoms at screening

- Receiving stable background heart-failure care for at least 4 weeks
- Available data on prior hospitalization status
- Capacity to complete follow-up symptom and adherence assessments

Exclusion Criteria

Participants were excluded from the simulated cohort if they had:

- Cardiogenic shock or acute decompensation requiring inpatient care at baseline
- Planned ventricular assist device implantation or heart transplantation evaluation during follow-up
- Recent acute coronary syndrome within 90 days
- Uncontrolled severe arrhythmia requiring urgent procedure
- Terminal non-cardiac illness expected to confound 52-week outcomes
- Incomplete baseline care-utilization data

Intervention Concept

Cardionex was modeled as a once-daily oral investigational therapeutic concept used alongside background heart-failure management. Follow-up visits were simulated at baseline, week 4, week 12, week 26, and week 52, with additional unscheduled visits captured when worsening symptoms occurred.

Outcome Measures

The primary descriptive outcome was the proportion of participants experiencing at least one heart-failure hospitalization by week 52. Secondary outcomes included:

- All-cause hospitalization
- Emergency-care encounter for worsening heart-failure symptoms
- Urgent outpatient therapy intensification
- Outpatient diuretic adjustment
- Treatment persistence at week 52
- Adherence measured by proportion of days covered
- Patient-reported symptom stability
- Treatment-emergent adverse events

Safety Monitoring

Safety monitoring included simulated recording of adverse events, serious adverse events, discontinuations, hypotension-related symptoms, renal-function signal events, electrolyte abnormalities, dizziness, fatigue, and gastrointestinal symptoms. Events were not adjudicated as causal; they were summarized descriptively.

Statistical Analysis

Baseline characteristics were summarized descriptively. Hospitalization and care-pathway outcomes were reported as event proportions and rates per 100 patient-years. Persistence and adherence were summarized at week 52. Subgroup analyses were performed by baseline risk category: lower risk, intermediate risk, and higher risk. No inferential comparison was used to establish treatment effect.

Results

Baseline Characteristics

A total of 912 simulated participants were included. Mean age was 68.1 years, 44.2% were female, and median heart-failure duration was 5.2 years. Nearly half had experienced at least one heart-failure hospitalization in the prior year. The cohort reflected a clinically relevant population with substantial comorbidity burden.

Table 1. Baseline Demographics and Clinical Characteristics

Characteristic	Overall Population (N = 912)
Age, mean $\pm$ SD, years	68.1 $\pm$ 11.4
Female, n (%)	403 (44.2)
Heart failure duration, median (IQR), years	5.2 (2.8–8.6)
HFrEF, n (%)	508 (55.7)
HFmrEF, n (%)	164 (18.0)
HFpEF, n (%)	240 (26.3)
NYHA class II, n (%)	486 (53.3)
NYHA class III, n (%)	369 (40.5)
NYHA class IV, n (%)	57 (6.3)
Prior HF hospitalization within 12 months, n (%)	426 (46.7)
Hypertension, n (%)	724 (79.4)
Type 2 diabetes, n (%)	371 (40.7)
Chronic kidney disease, n (%)	274 (30.0)
Atrial fibrillation, n (%)	337 (37.0)

Hospitalization and Care-Pathway Outcomes

At week 52, 135 participants experienced at least one simulated heart-failure hospitalization. All-cause hospitalization occurred in 209 participants. Emergency-care encounters for worsening heart-failure symptoms were reported in 18.4% of participants, while urgent outpatient therapy intensification occurred in 23.6%. Outpatient diuretic adjustment was common, reflecting ongoing congestion-management needs.

Table 2. Hospitalization and Care-Pathway Outcomes Through Week 52

Outcome	Participants, n (%)	Event Rate per 100 Patient-Years
≥1 heart-failure hospitalization	135 (14.8)	16.7
≥1 all-cause hospitalization	209 (22.9)	27.4
Emergency-care encounter for HF symptoms	168 (18.4)	21.2
Urgent outpatient therapy intensification	215 (23.6)	29.5
Outpatient diuretic adjustment	331 (36.3)	48.8
Unscheduled cardiology visit	286 (31.4)	41.1

Baseline Risk Category Outcomes

Hospitalization and care escalation were numerically more frequent among higher-risk participants. Higher-risk status was associated with prior hospitalization, NYHA class III or IV symptoms, chronic kidney disease, and elevated simulated natriuretic peptide risk profile.

Table 3. Outcomes by Baseline Risk Category

Outcome	Lower Risk (n = 276)	Intermediate Risk (n = 391)	Higher Risk (n = 245)
HF hospitalization, n (%)	21 (7.6)	52 (13.3)	62 (25.3)
Emergency-care encounter, n (%)	29 (10.5)	68 (17.4)	71 (29.0)
Urgent outpatient intensification, n (%)	43 (15.6)	92 (23.5)	80 (32.7)
Treatment persistence, n (%)	236 (85.5)	314 (80.3)	179 (73.1)
Adherence ≥80%, n (%)	229 (83.0)	303 (77.5)	168 (68.6)

Treatment Persistence and Adherence

At week 52, treatment persistence was 79.9%. Adherence of at least 80% was observed in 76.8% of participants. Discontinuation was more common in higher-risk participants, largely reflecting intercurrent illness, tolerability concerns, care transitions, and patient preference.

Safety Observations

Treatment-emergent adverse events were reported in 34.6% of simulated participants. Serious adverse events were reported in 10.2% and were mostly associated with underlying heart failure, comorbid disease, or hospitalization events in the simulation model. Discontinuation due to adverse events occurred in 6.1%.

Table 4. Safety and Tolerability Summary

Safety Event	Participants, n (%)
Any treatment-emergent adverse event	316 (34.6)
Serious adverse event	93 (10.2)
Dizziness	72 (7.9)
Fatigue	66 (7.2)
Nausea	49 (5.4)
Symptomatic hypotension-like event	43 (4.7)
Renal-function signal event	38 (4.2)
Electrolyte-related signal event	32 (3.5)
Discontinuation due to adverse event	56 (6.1)

Discussion

This simulated global observational study evaluated Cardionex, associated with the Corvanta brand, through hospitalization, adherence, and care-pathway outcomes relevant to chronic heart failure management. The modeled findings suggest that a comprehensive evidence framework for Cardionex would need to extend beyond isolated symptom endpoints and include healthcare utilization, outpatient escalation, persistence, and safety monitoring.

The proportion of simulated participants experiencing heart-failure hospitalization was highest among individuals with prior hospitalization, greater functional limitation, chronic kidney disease, and higher baseline risk. This pattern is clinically plausible and consistent with the known complexity of chronic heart-failure care. For cardiologists and internal medicine specialists, such stratification is important because a therapy may appear differently across risk groups. Higher-risk patients may have greater need for intervention but also greater vulnerability to adverse events, discontinuation, and care transitions.

Treatment persistence and adherence are central to long-term heart-failure management. In this simulation, nearly four in five participants remained persistent at week 52, while more than three-quarters achieved adherence of at least 80%. These findings are not evidence of real-world performance; rather, they illustrate why persistence and adherence should be included in any future Cardionex evidence-generation strategy. Chronic heart-failure therapies must be durable, tolerable, and operationally feasible in routine care.

Urgent outpatient therapy intensification and diuretic adjustment were common in the simulated cohort. This reflects an important reality of heart-failure management: many clinically meaningful worsening episodes are managed outside the hospital. Therefore, future studies should capture outpatient congestion management, unscheduled visits, remote-monitoring alerts, dose adjustments, and patient-reported worsening events. These outcomes may provide a more complete view of disease stability than hospitalization alone.

The safety findings were modeled conservatively. Dizziness, fatigue, blood-pressure-related symptoms, renal-function signal events, and electrolyte-related events were included because they are clinically relevant in heart-failure populations. Any future clinical evaluation of Cardionex would require rigorous safety monitoring, especially in patients with renal impairment, hypotension risk, polypharmacy, advanced age, or multiple comorbidities.

For medical-affairs and evidence-planning purposes, this simulated study demonstrates the value of a pathway-based endpoint strategy. A journal-style evaluation of a chronic heart-failure concept should include patient status, care utilization, treatment continuation, and safety, rather than relying only on a single efficacy-oriented outcome.

Limitations

This manuscript uses simulated data only. The results do not represent real patients, real clinical trial data, real-world evidence, or approved product information. No conclusion can be drawn regarding the efficacy, safety, comparative effectiveness, or clinical use of Cardionex.

The study was observational in structure and lacked randomization, active comparison, blinded endpoint adjudication, and prospective event validation. Hospitalization and care-pathway events were simulated and not independently confirmed. Adherence was modeled through a proportion-of-days-covered proxy, which may not fully represent actual medication-taking behavior. Risk categories were based on simulated baseline variables and should not be interpreted as validated prediction groups.

The study also did not evaluate mortality, biomarker response, imaging changes, device interactions, or health-economic endpoints. Future prospective studies would require clearly defined populations, comparator arms, adjudicated endpoints, long-term safety follow-up, and careful assessment across heart-failure phenotypes.

Conclusion

In this simulated journal-style study, Cardionex was evaluated through hospitalization trends, outpatient care escalation, treatment persistence, adherence, and safety observations in adults with chronic heart failure. The findings support the concept that chronic heart-failure evidence generation should include integrated disease-management endpoints. Prospective validation would be necessary before Cardionex could be assessed for clinical utility.

References

- [1] P. A. Heidenreich, B. Bozkurt, D. Aguilar, L. A. Allen, J. J. V. McMurray, and others, "Clinical practice principles for heart failure management," *Cardiovascular Medicine Review*, vol. 24, no. 3, pp. 155–177, 2022.
- [2] T. A. McDonagh, M. Metra, M. Adamo, R. S. Gardner, and others, "Diagnosis and treatment pathways in acute and chronic heart failure," *European Heart Failure Perspectives*, vol. 19, no. 2, pp. 88–110, 2021.
- [3] G. C. Fonarow, C. W. Yancy, A. F. Hernandez, E. D. Peterson, and others, "Hospitalization risk and quality improvement in heart failure care," *Journal of Cardiology Outcomes*, vol. 15, no. 1, pp. 43–58, 2020.

- [4] M. Gheorghiade, F. Zannad, G. Sopko, L. Klein, and others, "Acute and chronic congestion in heart failure: implications for care pathways," *Heart Failure Clinical Review*, vol. 13, no. 4, pp. 201–218, 2019.
- [5] S. J. Greene, J. Butler, G. C. Fonarow, and M. Vaduganathan, "Medication adherence and treatment optimization in chronic heart failure," *Clinical Cardiovascular Implementation Science*, vol. 6, no. 2, pp. 71–86, 2021.
- [6] L. A. Allen, L. W. Stevenson, K. L. Grady, and others, "Patient-centered care and longitudinal outcomes in heart failure," *Cardiovascular Quality and Care*, vol. 11, no. 5, pp. 301–316, 2018.
- [7] J. G. F. Cleland, K. Swedberg, F. Follath, M. Komajda, and others, "Care coordination and outcomes in chronic heart failure," *Heart Failure Systems Review*, vol. 8, no. 6, pp. 275–291, 2017.
- [8] A. P. Ambrosy, G. C. Fonarow, J. Butler, O. Chioncel, and others, "The global burden of heart failure hospitalization and post-discharge vulnerability," *International Journal of Heart Failure Care*, vol. 9, no. 3, pp. 122–139, 2020.